Harshit Sharma

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Portfolio: harshit-ai-portfolio.vercel.app

Professional Summary

Final-year **AI/ML Engineer** specializing in **Computer Vision, RAG Systems, and AI Agent Development**. Hands-on experience building real-time anomaly detection platforms, hybrid retrieval pipelines, and open-vocabulary vision systems. Strong expertise in **LLMs, Vector Search, Docker, Kubernetes, and MLOps workflows**. **Microsoft Hackathon Finalist** with proven ability to ship production-grade AI systems end-to-end.

Technical Skills

Programming Languages: Python, SQL, C++, JavaScript, TypeScript
AI/LLM Engineering: LangChain, RAG Pipelines, Hybrid Retrieval, Semantic Search, Embeddings, FAISS, Cross-Encoder Rerankers, Hugging Face Transformers, AI Agents, Multi-Agent Systems
Computer Vision: YOLOv8, YOLO-World, Grounding DINO, OpenCV, Real-time Video Analytics, Open-Vocabulary Detection
Machine Learning: Deep Learning, Model Benchmarking, Feature Engineering, Hyperparameter Tuning, Model Optimization, Supervised & Unsupervised Learning
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy
Backend & APIs: NestJS, Flask, REST APIs, MongoDB, MySQL, n8n Workflow Automation
DevOps & Deployment: Docker, Kubernetes, Git, GitHub, CI/CD, Linux, Cloud Deployment
Frontend & Tools: React, Electron, Postman, VS Code
Data Analytics & Visualization: Power BI, Excel, Matplotlib, Seaborn, EDA, Statistical Analysis

Professional Experience

AI Developer Intern — **Dec 2025 – May 2026**
Codeacious Technologies — *On-site*

- Designed and deployed a real-time visual anomaly detection platform that monitors live video streams and automatically flags unusual activities, removing the need for manual surveillance review
- Built an open-vocabulary detection pipeline that identifies anomalies described in natural language prompts, eliminating the need to retrain models for new threat scenarios
- Contributed to a NemoClaw-based forked repository by building an Electron desktop app in TypeScript with sandboxed multi-LLM agents integrated with Slack, Notion, and GitHub
- Developed a production-grade RAG pipeline enabling natural-language querying over large document collections with accurate, source-grounded responses
- Implemented a reranking layer over vector similarity search to filter out semantically close but irrelevant retrievals, significantly boosting answer precision
- Built the end-to-end backend on NestJS and MongoDB, paired with a React dashboard for real-time monitoring, alert management, and analytics
- Containerized the AI stack with Docker and orchestrated services via Kubernetes for scalable, fault-tolerant deployments under high traffic loads
- Automated alert escalation and downstream integrations through n8n workflows to streamline notification routing across connected systems
- Benchmarked YOLOv8, YOLO-World, and Grounding DINO to optimize trade-offs between latency, detection accuracy, and open-vocabulary flexibility

Key Projects

Event Anomaly Detection System — Real-time Streaming | *YOLOv8, YOLO-World, Grounding DINO* — **2025**

- Built a production-grade surveillance intelligence platform on the Clarity framework that monitors live video streams and detects abnormal events in real time
- Enabled open-vocabulary detection so users can define new anomaly categories in plain language without any model retraining
- Integrated YOLOv8, YOLO-World, and Grounding DINO into the detection pipeline, selecting the right model per use case based on speed and accuracy needs
- Used Clarity for video ingestion and streaming, MongoDB for event storage, and a React dashboard for live monitoring; alerts routed through n8n
- **Tech Stack**: Clarity, YOLOv8, YOLO-World, Grounding DINO, OpenCV, NestJS, React, MongoDB, n8n

Advanced RAG Pipeline with Reranking & Hybrid Retrieval | *FAISS, LangChain, Transformers* **2025**

- Architected a RAG system that converts unstructured document collections into queryable, citation-backed knowledge bases, grounding every LLM response in real source material
- Experimented with multiple retrieval strategies including dense vector search via FAISS, sparse keyword-based retrieval, and hybrid approaches combining both
- Implemented a dedicated reranker (RR) that re-scores candidate chunks against the query before passing the most relevant ones to the LLM, sharply reducing irrelevant context
- Applied semantic chunking to preserve cross-passage context, ensuring retrieved passages stay coherent rather than fragmenting mid-thought
- **Tech Stack**: FAISS, Hugging Face Transformers, LangChain, Cross-Encoder Rerankers, Hybrid Retrieval, Python

Crime Data Classification System | *ML, SMOTE, Feature Engineering* **2025**

- Developed ensemble classification models (Logistic Regression, Random Forest, XGBoost) to predict crime categories from historical incident data
- Tackled severe class imbalance using SMOTE oversampling combined with feature scaling, significantly improving minority-class recall
- Performed exploratory data analysis with visualizations to surface temporal and geographic crime patterns that informed feature design
- **Tech Stack**: Scikit-learn, XGBoost, SMOTE, Pandas, Matplotlib, Seaborn

Portfolio Optimization Using Deep Learning | *ANNs, Financial Modeling* **2025**

- Implemented Artificial Neural Networks to learn optimal asset allocation strategies directly from historical market data
- Applied deep learning to predict portfolio weights that balance expected return against downside risk, capturing non-linear market behavior
- **Tech Stack**: TensorFlow, Keras, Pandas, NumPy, Financial APIs

Education

Gautam Buddha University **Greater Noida, India**
*Bachelor of Technology in Information Technology — **CGPA: 8.48/10*** *Expected: 2026*

- **Relevant Coursework**: Machine Learning, Deep Learning, Data Structures, Algorithms, Database Systems, Statistics

Certifications & Achievements

Certifications: SQL (HackerRank), Python (Kaggle), Pandas (Kaggle), Power BI, Advanced Excel, Machine Learning A-Z, Statistics for Data Science
Achievements: Finalist — Hackathons Ignite 2025 organized by Microsoft at Gautam Buddha University

Additional Information

Languages: English (Fluent), Hindi (Native)
Interests: Computer Vision Research, Open-Vocabulary Detection, AI Agents, Semantic Search, MLOps, AI Safety, Hackathons